

A Level-Set Framework for Multilayer Aircraft Ice Accretion

Abstract

This work presents a continuous level-set methodology for the prediction of multilayer aircraft ice accretion on unstructured meshes. Conventional multilayer icing simulations commonly update the ice geometry through explicit displacement of surface nodes, which can become sensitive to mesh quality as strongly curved and interacting ice features develop. The proposed methodology instead represents the evolving ice–air interface implicitly as the zero contour of a signed-distance level-set function. A key contribution of this work is the extension of the continuous level-set formulation to the prediction of complex 3D multilayer ice-accretion geometries. Implemented within the CHAMPS icing framework, the methodology combines surface-quantity propagation, semi-implicit Rosenbrock–Wanner (ROW) time integration, and conformal surface extraction using MMG. The methodology is assessed for representative 2D and 3D configurations from the AIAA Ice Prediction Workshops (IPW). The results demonstrate robust multilayer ice accretion in both 2D and 3D, with increasing numbers of accretion layers showing convergence of the predicted ice geometries toward experimental measurements.

Introduction

Aircraft ice accretion can significantly degrade aerodynamic performance by modifying the geometry of lifting surfaces [1]. Numerical icing simulations generally employ a quasi-steady multilayer approach in which aerodynamic, droplet-impingement, and thermodynamic solutions are successively recomputed as the ice geometry evolves [2].

Geometry evolution remains an important numerical challenge. Conventional methods typically displace surface nodes along the local surface-normal direction [3]. Although computationally efficient, repeated normal displacement can lead to mesh distortion or entanglement as strongly curved or interacting ice features develop, particularly in 3D [4, 5].

Level-set methods represent evolving interfaces implicitly through a scalar field and provide a framework for tracking complex moving boundaries [6–8]. They have been applied to moving-interface problems and, more recently, to aircraft ice accretion [9–12].

The present work develops a continuous level-set framework for 2D and 3D multilayer ice accretion within the CHAMPS solver [13]. The significant aspect of the proposed approach is the evolution of the ice interface independently of the connectivity and displacement of the surface mesh, followed by conformal extraction and remeshing between successive accretion layers. This formulation is intended to

improve robustness for complex 3D ice geometries and interacting ice fronts.

Methodology

The methodology is embedded within the quasi-steady icing framework shown in Fig. 1. For each layer, the aerodynamic flow, droplet impingement, and surface thermodynamics are computed sequentially. The resulting ice thickness drives the level-set evolution, after which the new surface is extracted and remeshed for the subsequent layer.

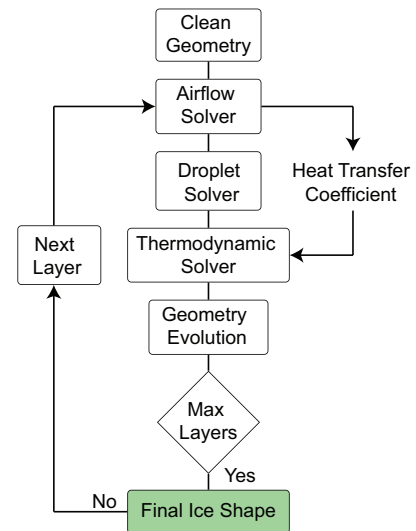


Figure 1. Quasi-steady multilayer ice-accretion framework implemented within CHAMPS.

Interface Representation and Advection

The ice–air interface is represented by a level-set function ϕ , with $\phi < 0$ in the ice, $\phi > 0$ in the fluid, and $\phi = 0$ at the interface, as illustrated in Fig. 2.

The initial signed-distance field, illustrated in Fig. 3(a), is obtained from the Eikonal equation [14],

$$|\nabla\phi| = 1, \quad \mathbf{n}_\phi = \frac{\nabla\phi}{|\nabla\phi|}, \quad (1)$$

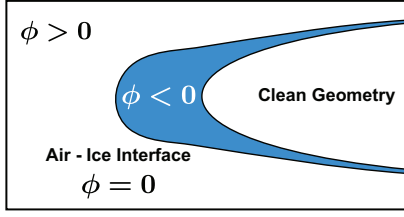


Figure 2. Level-set representation of the ice-air interface.

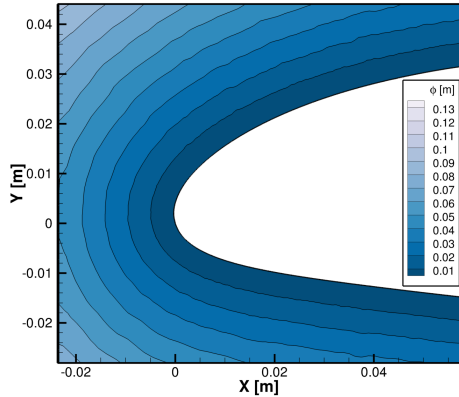
where $\mathbf{n}_\phi$ denotes the unit normal vector to the level-set contours. The interface is advected using the level-set equation [7],

$$\frac{\partial \phi}{\partial t} + \mathbf{V}_{ice} \cdot \nabla \phi = 0, \quad (2)$$

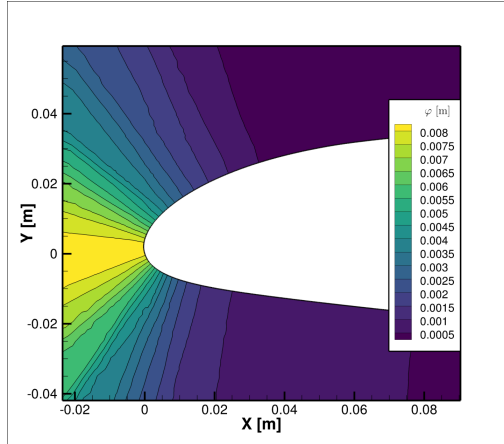
where the ice-growth velocity is

$$\mathbf{V}_{ice} = \frac{h_{ice}}{\Delta t_{layer}} \mathbf{n}_\phi. \quad (3)$$

Here, h_{ice} is the local ice thickness and Δt_{layer} is the accretion-layer duration. As illustrated in Fig. 3(b), the surface ice thickness is propagated into the level-set domain along $\mathbf{n}_\phi$ prior to level-set advection.



(a) Signed-distance initialization



(b) Ice-thickness propagation

Figure 3. Level-set initialization and ice-thickness propagation prior to interface advection.

Time Integration and Surface Extraction

To alleviate restrictive time-step limitations associated with explicit level-set integration, the level-set equation is advanced using a two-stage semi-implicit Rosenbrock–Wanner (ROW) scheme [15]. The time integration is written as

$$\phi^{n+1} = \phi^n + \sum_{j=1}^s m_j S_j, \quad (4)$$

where the stage increments S_i are obtained from

$$\left(\frac{\mathbf{I}}{\gamma \Delta t} + \mathbf{J} \right)^n S_i = -R \left(\phi^n + \sum_{j=1}^{i-1} a_{ij} S_j \right). \quad (5)$$

Here, $R(\phi)$ is the spatial residual, $\mathbf{J} = \partial R / \partial \phi$ its Jacobian, and S_i the stage increments. The resulting linear systems are solved using the Generalized Minimal Residual (GMRES) method [16]. The semi-implicit formulation improves stability and permits larger time steps on the locally refined auxiliary mesh used for level-set advection, illustrated in Fig. 4. Following level-set advection, the $\phi = 0$ interface is extracted using MMG [17] and remeshed for the subsequent accretion layer.

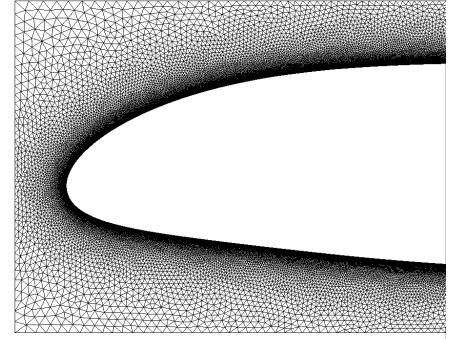


Figure 4. Adapted auxiliary mesh with local refinement in the ice-accretion region for level-set advection.

Preliminary

The methodology is first illustrated using the manufactured three-horn configuration introduced in [18]. This configuration features three distinct ice horns, producing highly concave and convex regions that challenge interface-tracking methods. An ice thickness of 0.02 m is uniformly imposed on all surface nodes with $x < 0.05$ m. As shown in Fig. 5, direct surface displacement can lead to mesh entanglement as neighboring fronts interact, whereas the implicit level-set representation naturally accommodates their evolution and merging.

The methodology is subsequently assessed using two configurations from the AIAA Ice Prediction Workshops [19, 20]. The corresponding icing conditions are summarized in Table 1.

2D Multilayer Convergence

IPW1 Case241 corresponds to rime ice accretion on a NACA23012 airfoil and is used to assess the convergence of the multilayer formulation [19]. Figure 6 compares the predicted ice geometries using 1, 10, and 20 accretion layers with the experimental ice shape.

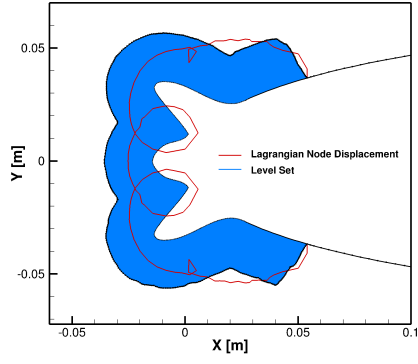


Figure 5. Evolution of interacting fronts using surface displacement and the level-set formulation.

Table 1. Summary of the IPW benchmark cases considered in this study.

Parameter	Case 241	Case 362
Geometry	2D NACA23012	3D swept NACA0012
Icing time (min)	5.0	20.0
LWC (g/m ³)	0.42	0.50
MVD (μ m)	30.0	34.7
M_∞	0.32	0.31
α (°)	2.0	0.0
C_{ref} (in.)	18	36
T_∞ (K)	250.15	266.0
Re	3.8×10^6	6.7×10^6

The experimental maximum ice height is 10.87 mm. The predicted maximum ice height increases from 10.16 mm using one layer to 10.69 and 10.79 mm using 10 and 20 layers, respectively. The corresponding relative difference with respect to experiment decreases from 6.47% to 1.68% and 0.70%. The diminishing difference between the 10- and 20-layer predictions indicates convergence of the predicted ice geometry with increasing temporal resolution.

3D Multilayer Ice Accretion

The 3D capability of the methodology is assessed using Case362, a swept NACA0012 configuration with a 36-in. chord and a 30° sweep angle [21]. The configuration exhibits spanwise-varying leading-edge ice characteristic of swept-wing icing. The level-set method remains

robust throughout the multilayer process, successively advecting, extracting, and remeshing the 3D interface. A 3D view of the predicted ice geometry is shown in Fig. 7.

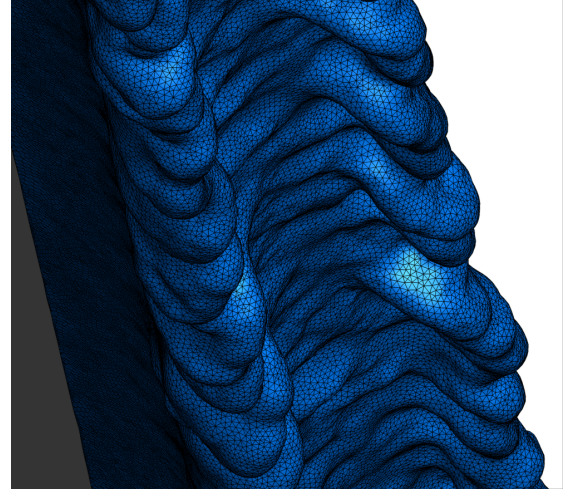


Figure 7. 3D view of the predicted ice geometry for Case 362.

The predicted ice geometry shows good agreement with the experimental ice shape and remains within the IPW2 participant envelope [20], as shown in Fig. 8.

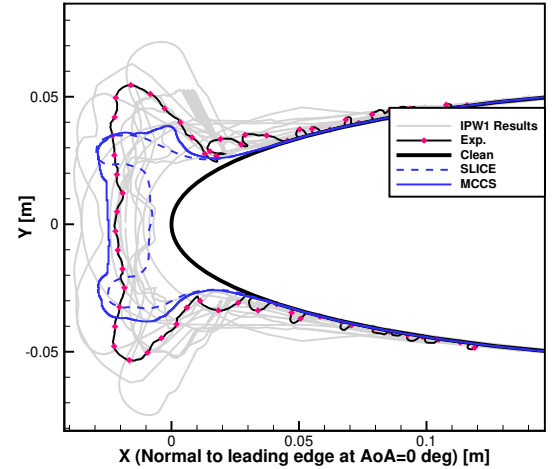


Figure 8. Comparison of the predicted Case 362 ice geometry with experimental and IPW2 participant data.

Expected Results

The completed study will extend the assessment of the level-set methodology to additional configurations from IPW2, with emphasis on complex 3D glaze-ice geometries. These cases will provide further evaluation of the robustness of the interface-propagation, extraction, and remeshing procedure over successive accretion layers. The resulting predictions will be compared with available experimental measurements and workshop participant data to assess ice-shape agreement.

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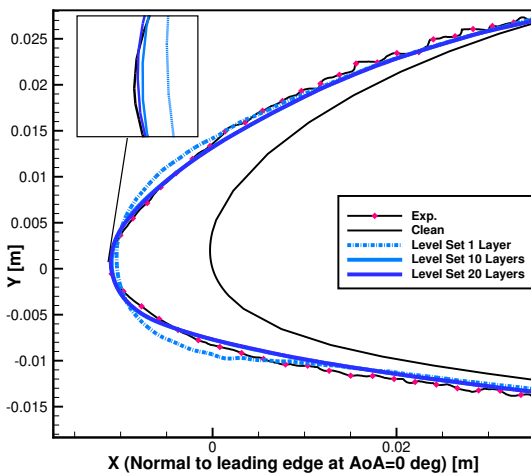


Figure 6. IPW1 Case 241 predictions using 1, 10, and 20 accretion layers.

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Definitions, Acronyms, Abbreviations

GMRES	Generalized Minimal Residual
IPW	AIAA Ice Prediction Workshop
RANS	Reynolds-Averaged Navier–Stokes
ROW	Rosenbrock–Wanner

General symbols

h_{ice}	Local ice thickness	[m]
$\mathbf{n}_\phi$	Level-set interface normal	[–]
Δt_{layer}	Accretion-layer duration	[s]
$\mathbf{V}_{ice}$	Ice-growth velocity	[m/s]

Greek symbols

ϕ	Level-set signed-distance function	[m]
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